



Diethyl ether cracking and ethanol dehydration: Acid catalysis and reaction paths

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HIGHLIGHTS

- H-zeolites are more active than alumina and silica–alumina in ethanol dehydration.
- Reaction of ethoxy-groups with undissociated ethanol produces diethylether (DEE).
- H-zeolites, alumina and silica–alumina are active in DEE cracking and hydrolysis.
- Ethylene is produced by cracking of both DEE (low *T*) and ethoxy-groups (high *T*).

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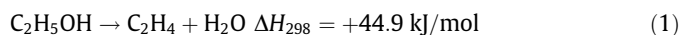
ABSTRACT

The catalytic conversion of ethanol and diethyl ether (DEE) was studied over alumina, zeolites MFI, FER and USY, silica–alumina and calcined hydrotalcite. Zeolites, alumina and silica–alumina are active in the temperature range 453–573 K for both ethanol dehydration to DEE and to ethylene and for DEE cracking and hydrolysis, producing back ethanol and ethylene. Protonic zeolites are more active than alumina which is slightly more active than silica–alumina for these reactions. Ethanol dehydration to DEE occurs selectively at lower temperature with a “bimolecular” mechanism involving reaction of ethoxy groups with undissociated ethanol. Ethanol dehydration to ethylene occurs selectively at relatively high temperature with an elimination mechanism via decomposition of ethoxy groups over these catalysts, but also occurs, at lower temperature, with a consecutive path via DEE formation and cracking.

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1. Introduction

In the frame of a future possible organic chemistry based on renewables, bioethanol coming from lignocellulosics could play an important role as a primary intermediate [1]. The production of ethylene from bioethanol as a secondary intermediate can allow the further synthesis of polyethylene and of a large number of tertiary intermediates, produced with the current petrochemical technologies, but arising from renewable feedstock. Ethanol can be converted into ethylene by catalytic dehydration:



The reaction (1) is endothermic, but is largely favored thermodynamically already at moderate temperatures (e.g. 473–573 K). This reaction has indeed been applied at the industrial level in the sixties using aluminas as the catalysts [2,3]. On the other hand, a number of studies reported on the high catalytic activities

of different zeolite catalysts [4–7]. The literature in the field has been recently reviewed by Zhang and Yu [8], that concluded that zeolites might be unstable for this reaction. In contrast, Fan et al. judged they are applicable at the industrial level to produce ethylene from bioethanol [9].

Reaction (1) suffers somehow of the competition with the production of diethyl ether (DEE):



which is exothermic, also favored at low or moderate temperature. Over most catalysts, reaction (2) occurs very selectively at low temperature and conversion while reaction (1) occurs at higher temperature and very high conversion. Different opinions are reported concerning the path involved in these reactions. The reaction (1) performed in the liquid phase with concentrated sulfuric acid as a catalyst at ca. 450 K is supposed to occur with a “monomolecular” mechanism, either with an E_2 (bimolecular elimination) concerted elimination mechanism, while reaction (2) occurring at lower temperature (410 K) is reported to occur with a S_{N}^2 (bimolecular nucleophilic substitution) bimolecular mechanism from a protonated

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form of ethanol and an non-protonated one [10,11]. According to DFA (density functional analysis) computational studies, the E₂ mechanism should be the most favored also on alumina for reaction (1) [12]. Roca et al. working on silica alumina concluded that DEE and ethylene are produced by parallel reactions [13]. Also DeWilde et al. concluded that the bimolecular mechanism producing DEE and the unimolecular mechanism producing ethylene are parallel on alumina, both being inhibited by water [14]. According to these authors, two ethanol undissociatively adsorbed “dimer” species react together forming DEE, while ethylene is formed by decomposition of ethoxy- groups. Other authors believe that DEE can be formed by reaction of an adsorbed ethanol molecule with an ethoxide group [15] or of two ethoxide groups [16]. Several authors [12,17–22] also suggest that diethyl ether might be an intermediate in the dehydration of ethanol to ethylene.

The selective production of DEE can be of interest because it is in fact a considerable product of the chemical industry, mostly used as a solvent in a number of fine chemistry, fragrance and pharmaceutical chemical processes, and in some processes involved in explosives synthesis [23]. It finds also a number of applications in fuel chemistry [24] as an ignition improving additive in both diesel and gasoline engines according to its many useful properties: high volatility, cetane number higher than 125 [25] and octane number higher than 110 [26], reasonable energy density for on-board storage, high oxygen content, low auto-ignition temperature, broad flammability limits, and high miscibility in diesel fuel and bioethanol. Partial conversion of ethanol to DEE has been studied in order to increase volatility of ethanol to be applicable as a gasoline fuel also in cold start conditions [27]. Disadvantages include the same high volatility, propensity for peroxidation in storage, low lubricity and anesthetic effects [28].

Diethyl ether is currently either produced as a byproduct of ethanol synthesis by ethylene hydration processes [29] or by dehydration of (bio)ethanol either with sulfuric acid or in gas phase over alumina catalysts [30], thus forming a renewable chemical. The reverse reactions, i.e. the cracking of diethyl ether to ethylene and/or to ethanol can be of interest from several points of view: (i) to check the conclusion of some authors [17–20] that DEE acts as an intermediate in the ethylene synthesis from (bio)ethanol; (ii) to recover ethanol or/and ethylene from excess of DEE in the synthesis processes; (iii) to convert spent DEE into more useful products and to abate DEE vapors, which have some toxicity associated to its well-known anesthetic behavior; (iv) to model the conversion of natural and renewable substances containing ether bonds, such as e.g. lignin.

In the present paper, we will summarize the results of a screening study on the catalytic cracking of DEE, a correlation of the acido-basic properties of the catalysts with the catalytic behavior and some consideration on the reaction path concerning ethanol dehydration on acid catalysts.

2. Experimental

2.1. Catalysts

The properties of the catalysts, all commercial, are summarized in Table 1. SA is a commercial cracking catalyst of the silica-alumina type, MFI is a Si-rich form of H-ZSM-5 zeolite and USY (Ultra-Stable Y zeolite) is a dealuminated protonic faujasite. The composition of FER zeolite has a typical ferrierite composition.

2.2. Catalytic experiments

Catalytic experiments were performed at atmospheric pressure in a tubular flow reactor (i.d. 6 mm) using 0.5 g catalyst (60–70

mesh sieved, thus with a ratio between the particle diameter and the internal reactor diameter near 25) and feeding 71.6% v/v diethyl ether (DEE) in nitrogen with 10.42 h^{−1} weight hourly space velocity (WHSV) (total flow rate of 40 cc/min), in the case of ethanol feeding 7.9% v/v ethanol in nitrogen with 1.43 h^{−1} WHSV (total flow rate of 80 cc/min). The carrier gas (nitrogen) was passed through a bubbler containing DEE (Carlo Erba, 99.9%) or ethanol (Sigma–Aldrich, 96%) maintained at constant temperature (298 K) in order to obtain the desired partial pressures. The temperature in the experiment was varied stepwise from 423 K to 723 K for DEE and from 373 K to 623 K for ethanol.

In order to have more information about mechanism, the experiments were performed at atmospheric pressure in a tubular flow reactor (i.d. 6 mm) using 0.5 g catalyst (60–70 mesh sieved) and feeding the mixture of ethanol + water + diethyl ether. An ethanol–water–diethyl ether single phase liquid solution was pumped through a HPLC pump (Shimadzu Corp. LC-D10 AD) in an appropriately heated (573 K) section of the feed line to produce a gas whose composition corresponds to molecular ratio of DEE:water:EtOH = 5:5:3 at atmospheric pressure and 298 K. The helium carrier gas of 67 cc/min will bring the gas mixture of DEE (5 cc/min) + H₂O (5 cc/min) + EtOH (3 cc/min) to the reactor and the reaction temperature was varied stepwise from 423 K to 673 K.

Reactant (R) conversion is defined as usual:

$$X_R = (n_{R(in)} - n_{R(out)})/n_{R(in)}$$

While selectivity to product *i* is defined as follows:

$$S_i = n_i/(v_i(n_{R(in)} - n_{R(out)}))$$

where *n_i* is the moles number of compound *i*, and *v_i* is the ratio of stoichiometric reaction coefficients.

The outlet gases were analyzed by a gas chromatograph (GC) Agilent 4890 equipped with a Varian capillary column “Molsieve 5A/Porabond A Tandem” and TCD and FID detectors in series. In order to identify the compounds of the outlet gases, a gas chromatography coupled with mass spectroscopy (GC–MS) Thermo Scientific with TG–SQC column (15 m × 0.25 mm × 0.25 μm) was used.

2.3. Infrared spectroscopy (IR) experiments

IR spectra were recorded using Nicolet 380 FT-IR spectrometer. Acidity measurements were done using the pure powders pressed into thin wafers and activated in the IR cell connected with a conventional outgassing/gas-manipulation apparatus at 773 K. The activated samples were contacted with pyridine vapor (p_{Py} ~1 torr) at room temperature for 15 min; after which the IR spectra of the surface species were collected in continuous evacuation at room temperature with increasing temperature.

In order to study the mechanism of reaction, pressed disks of the pure catalyst powders were activated *in situ* in the IR cell connected with a conventional gas-manipulation apparatus before any adsorption experiment. IR spectra of the surface species as well as of the gas phase were collected upon increasing temperature in static conditions (starting p_{EtOH} ~4 torr).

3. Results

3.1. Thermodynamic considerations

The composition of the system starting from pure ethanol at atmospheric pressure, as expected in conditions of thermodynamic control was calculated using the Soave–Redlich–Kwong (SRK) equation [31] and using HSC Chemistry 5.11 program (Fig. 1). Above ca. 400 K ethylene and water are the only products

Table 1
The properties of investigated catalysts.

Notation	Phase	Composition	Commercial name	Manufacturer	Preparation	S_{BET}^a
MgA	Calcined hydrotalcite	MgO/Al ₂ O ₃ wt%70:30	Pural MG70	Sasol	Calcined at 773 K for 4 h	195 ± 10 ^b
A	Alumina	Al ₂ O ₃	Puralox SBA 200	Sasol	As received	190 ± 10
SA	Silica–alumina	SiO ₂ –Al ₂ O ₃ 13% wt Al ₂ O ₃	Cracking catalyst	Strem chemical	As received	330 ± 10
FER	H-ferrierite	SiO ₂ /Al ₂ O ₃ mol 20	CP 914C	Zeolyst	Calcined at 773 K, 4 h	400
MFI	H-ZSM-5	SiO ₂ /Al ₂ O ₃ mol 280	CBV 28014	Zeolyst	Calcined at 773 K, 4 h	400
USY	H-faujasite	SiO ₂ /Al ₂ O ₃ mol 30	CBV 720	Zeolyst	As received	780

^a From manufacturer.

^b BET measurement.

expected, while below 330 K diethyl ether and water are the most abundant products. DEE is more abundant than ethanol also in the range 330–400 K. Thus reaction (1) is the most favored above 400 K while reaction (2) is the most favored below 400 K.

3.2. Catalyst surface acidity characterization by IR spectroscopy of adsorption of pyridine

The surface acidity of the catalysts was characterized by IR spectroscopy of pyridine adsorption (Fig. 2). Over all catalysts bands at ca. 1602–1625, 1575, 1489 and 1440–1460 cm^{−1} are observed, due to the 8a, 8b, 19a and 19b modes of molecular pyridine [32], respectively. As it is well-known, the position of the 8a and 19b modes is indicative of the strength of the Lewis acid sites to which pyridine is coordinated. Bands due to pyridine bonded to strong Lewis acid sites (ν_{8a} ca. 1622 cm^{−1}, ν_{19b} > 1455 cm^{−1}) of the Al³⁺ type are well evident over A, SA and USY. Bands due to pyridine bonded to very weak Lewis sites (ν_{8a} ca. 1602 cm^{−1}) are observed on MgA while bands due to pyridine bonded to Lewis sites are not detected on MFI zeolite. Additional bands attributed to pyridinium ions (ν_{8a} ca. 1635 cm^{−1}, ν_{19a} at 1550–1535 cm^{−1}) are observed in the spectra of the zeolites as well as of SA, showing that these catalysts also have Brønsted acid sites [7]. The experiment concerning the FER zeolite is not reported because it is not really significant. In fact pyridine cannot enter in these conditions into the small cavities of ferrierite. Thus adsorbed pyridine only reveals external acidity in this case. Previous studies using smaller probe molecules such as acetonitrile showed the presence of strong Brønsted sites with some Lewis site also on the cavities of FER zeolite [33].

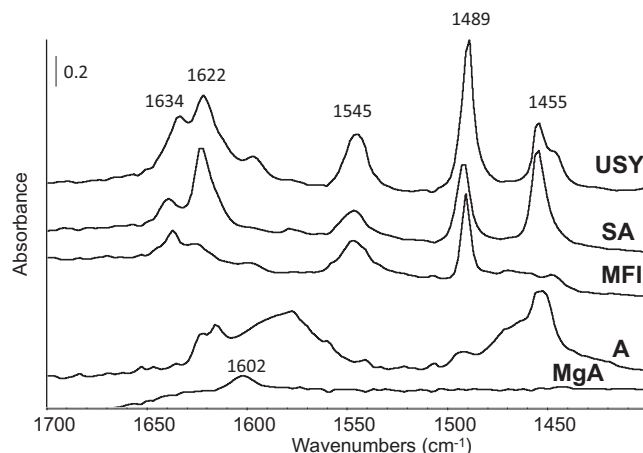


Fig. 2. FT-IR subtraction spectra of surface species arising from pyridine adsorbed on investigated catalysts after evacuated at 373 K.

3.3. Catalytic conversion of ethanol

The results of catalytic conversion experiments of ethanol over the catalysts under study are summarized in Fig. 3. The conversion of ethanol shows the following trend, e.g. at 473 K: FER (70.0%) > USY (62.9%) > MFI (44.4%) > A (20.8%) > SA (15.7%) > MgA (0.1%). This trend shows that zeolites are the most active in converting ethanol, due to their strong Brønsted acidity. A and SA catalysts have similar activity, attributed to Lewis acidity [34]. The very weakly acidic material MgA is very poorly active. In all cases, at low conversion, DEE is the largely predominant product, while at higher temperature and conversion, ethylene becomes the predominant product. However, on MgA a relevant amount of products different from ethylene and DEE is formed. They are mainly acetaldehyde and other carbonyl products providing evidence of the basic character of this catalyst.

To have more information, we performed ethanol conversion experiments by varying the contact time. The data on MFI are discussed here, because this catalyst was found to be a purely Brønsted acidic one. The ethanol conversion rate on MFI definitely depends on contact time, as shown in Fig. 4, i.e. the lower the space velocity (SV), the greater the conversion, as expected indeed. The data are reported as Table 2 in supplementary material. On the other hand, the DEE selectivity clearly has a reverse trend. At lower space velocities the DEE selectivity drops in favor of ethylene selectivity. To look deeper at this we considered the selectivities at the same conversion, at different SV and temperature. As for example, at conversions in the range 81–83%, the selectivity to DEE drops from 43.6% to 8.1% when temperature is increased from 523 K to 573 K in spite of the simultaneous strong increase in space velocity from 0.72 to 7.0 h^{−1}. This shows that reaction temperature has definitely a stronger effect, at high conversion levels, to favor ethylene selectivity, than conversion itself, i.e. ethanol partial pressure.

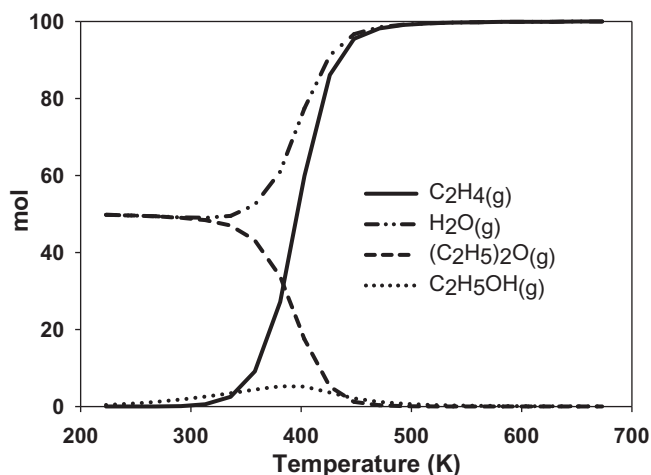


Fig. 1. Equilibrium compositions curve of ethanol, diethyl ether, ethylene and water in gas phase as a function of temperature.

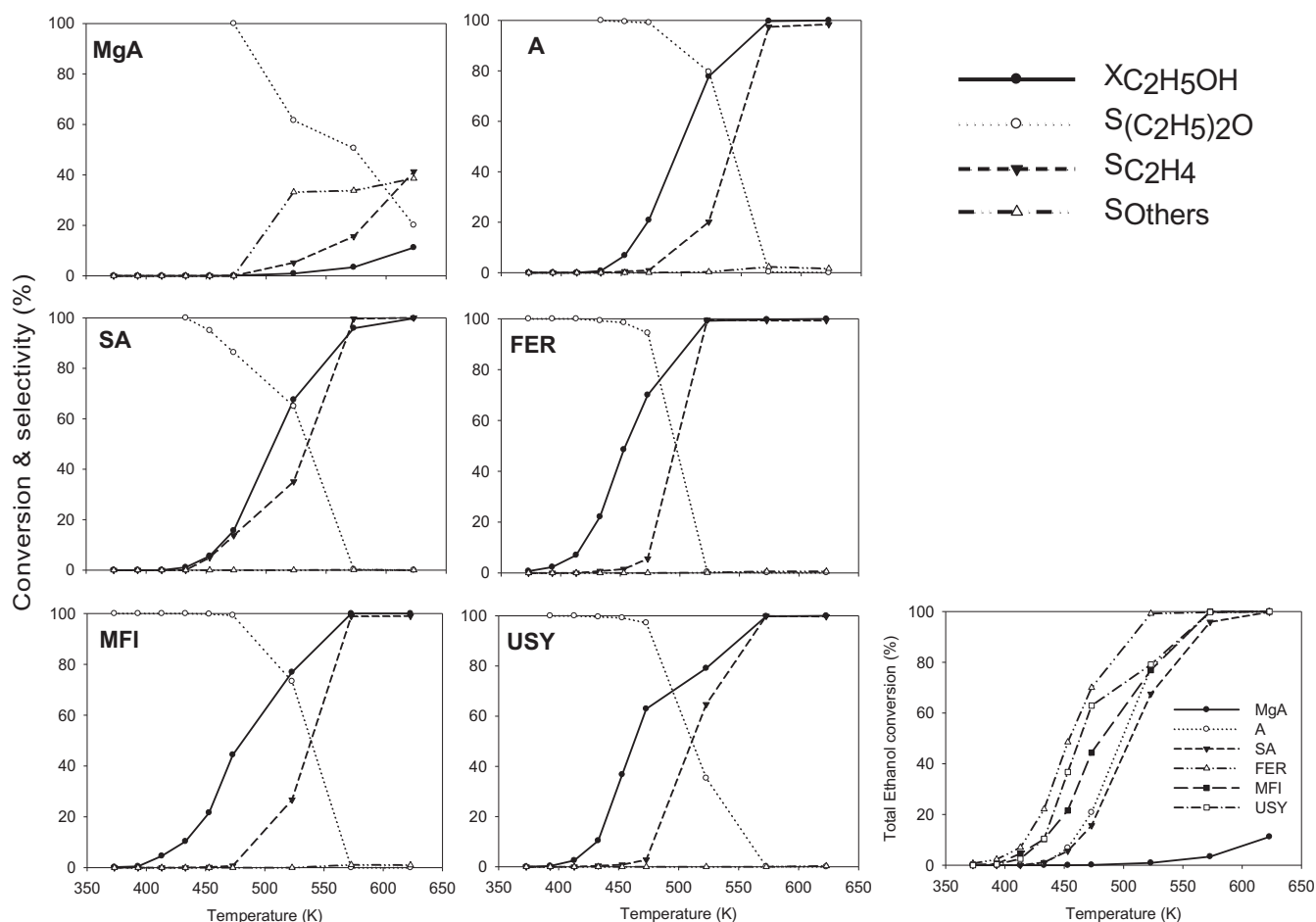


Fig. 3. Conversion and selectivities to C-containing products of ethanol over investigated catalysts at atmospheric pressure in a fixed-bed tubular quartz reactor using 0.5 g of catalysts with 1.43 h^{-1} WHSV in nitrogen, as a function of reaction temperature.

Thus the effect of the different activation energy of reaction (1) and (2) may be a discriminant factor.

At low space velocity and high temperature and conversion, ethylene selectivity is lowered by the production of higher hydrocarbons (Fig. 4). However, selectivities to higher hydrocarbons tend to vanish by increasing space velocity in favor of ethylene and even DEE selectivity, showing that these higher hydrocarbons are formed by over-conversion of ethylene.

Similar experiments were performed by varying space velocity on alumina, a purely Lewis acidic catalyst, with parallel results. In particular, also on alumina at lower space velocities the DEE selectivity drops in favor of ethylene selectivity.

3.4. Catalytic conversion of ethanol on alumina: IR experiments

IR experiments of ethanol adsorption and conversion were performed in order to have further information. The experiments performed on alumina (Fig. 5) show that ethanol adsorbs mainly in the form of ethoxy- groups, characterized mainly by the strong $\nu\text{C-O}/\nu\text{C-C}$ bands at 1116 and 1073 cm^{-1} , together with the CH deformation modes at 1450 and 1390 cm^{-1} and CH stretchings in the region $3000\text{--}2800 \text{ cm}^{-1}$. Small amounts of adsorbed undissociated ethanol (δCOH band at 1278 cm^{-1} , $\nu\text{C-O}/\nu\text{C-C}$ band at 1054 cm^{-1}) are also found. The spectra reported in Fig. 5 are those of the gas phase and of the adsorbed species detected after “saturating” the alumina surface with ethanol at room temperature and short evacuation of the line. After, temperature was progressively increased in static conditions. By heating up to at 423 K some

ethanol is released in the gas phase (bands at 1240 and 1066 cm^{-1}), while correspondingly the bands of undissociated adsorbed ethanol disappear. The bands of ethoxide species are essentially unchanged, thus showing that dissociative adsorption of ethanol is essentially irreversible in these conditions. Gas phase diethyl ether ($\nu\text{C-O}/\nu\text{C-C}$ band at 1142 cm^{-1}) is formed essentially in the step between 473 and 523 K , during which only traces of ethylene also form (CH_2 wagging mode at 949 cm^{-1}). It seems interesting to remark that upon this step the bands of ethoxy groups decrease significantly in intensity, and gas phase ethanol decreases significantly but not entirely. This clearly shows that ethoxy groups are indeed involved in the formation of DEE, and probably gas phase ethanol too. It seems also of interest to remark that adsorbed DEE is not found. Indeed parallel experiments, according to previous data [35], show that DEE adsorption is very weak, complete desorption being obtained by outgassing even at room temperature.

In the following step between 523 and 573 K , DEE gas disappears completely, and surface ethoxides too, almost completely. Also gas phase ethanol is no more observed, only ethylene being produced in big amounts ($\nu\text{CH}_2\text{s}$ at 3106 and 2988 cm^{-1} , ωCH_2 at 949 cm^{-1}). The IR experiments, formally consisting of steps at very long contact time, provide evidence of the strong effect of temperature on the reaction steps. It can be concluded that, in the step up to 473 K , the only phenomenon is the essentially irreversible adsorption and saturation of the surface with ethanol. In the range $473\text{--}523 \text{ K}$ the only fast reaction is the synthesis of DEE implying reaction of ethoxide species together with gas phase or weakly adsorbed ethanol. In spite of the very prolonged contact

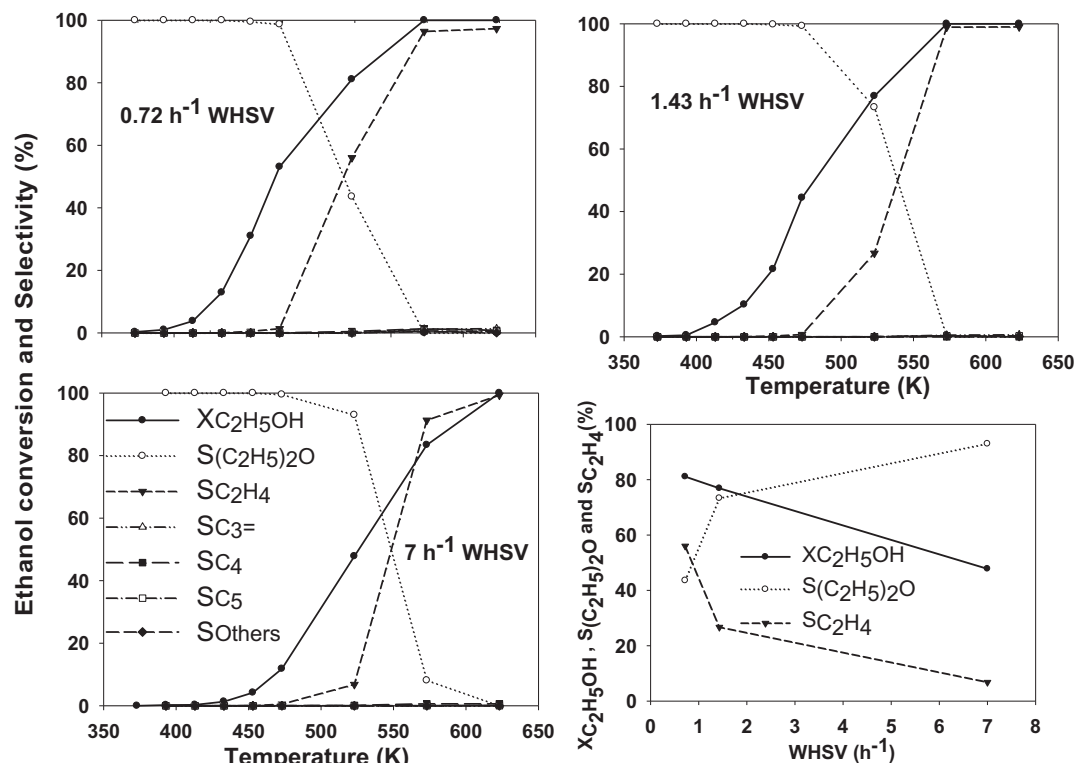


Fig. 4. Conversion and selectivities to C-containing products of ethanol over MFI at different space velocities; conversion, diethyl ether selectivity and ethylene selectivity on MFI at 523 K as a function of space velocity.

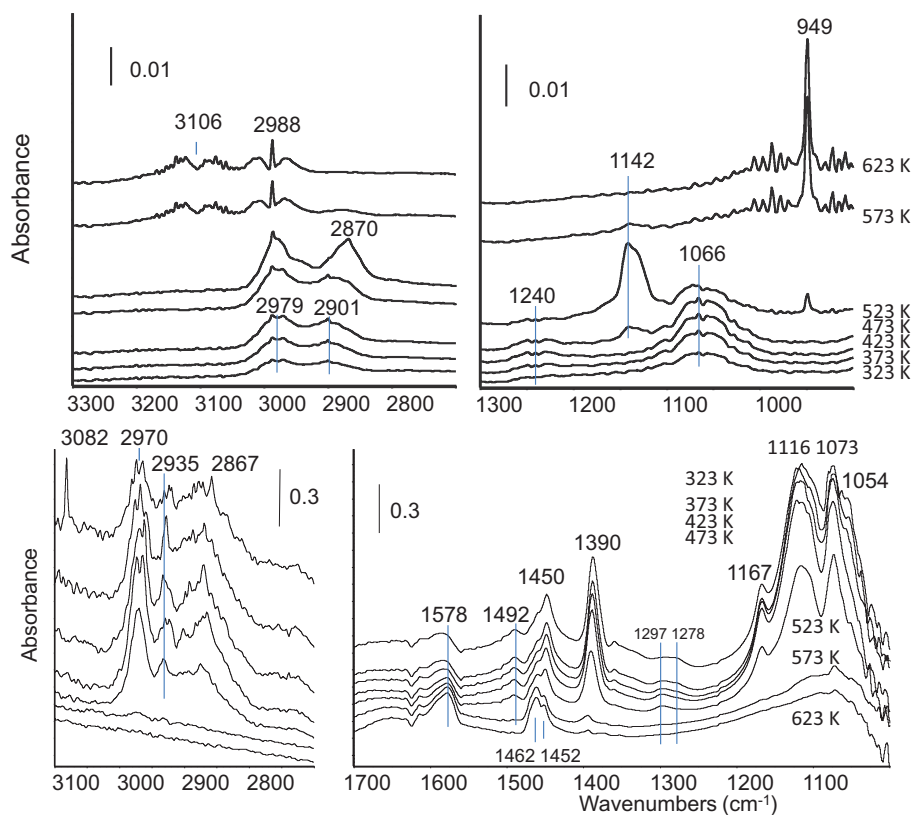


Fig. 5. FT-IR spectra of subtraction gas phase (top) and surface species (bottom) arising from ethanol adsorbed on alumina (A).

time, ethylene is essentially not formed. In the range 523–573 K both DEE cracking and ethoxy-group cracking become apparently fast, producing gas phase ethylene.

Similar experiments were performed with zeolite catalysts. The data obtained with FER are reported in Fig. 6. The IR spectra of the adsorbed species closely correspond to those reported for

H-mordenite by Kondo et al. [36] and attributed to a bridging ethoxy- group. The features of this alkoxy- species disappear in the 473–523 K range, when ethylene is produced as the only product. At lower temperature, alkoxy-groups coexist with gas phase ethanol and very small amounts of physisorbed ethanol. In this range, some DEE is formed in the gas phase while ethylene is not observed. These results support the idea that both on alumina and on FER, ethoxy-groups have two reactions paths: in the presence of gaseous or adsorbed ethanol, ethoxy- groups participate to the reaction producing DEE; when gas phase ethanol is no more available and at higher temperature, ethoxy- groups crack to ethylene.

3.5. Catalytic conversion of diethyl ether

In Fig. 7, the data concerning the DEE conversion over the investigated catalysts, are summarized. The conversion trend is, e.g. at 573 K, MFI (98.3%) > USY (83.8%) > A (55.9%) > SA (52.1%) > MgA (0.1%). Also in this case zeolites are evidently very active, while alumina and silica alumina have the diethyl ether conversion again similar but lower activity, and MgA has very low activity, confirming a scale based on acid strength governing the catalytic activity. Taking into account the apparent activity inversion between MFI and USY, it must be remarked that the activity of MFI is higher than that of USY also in ethanol dehydration at temperature lower than 473 K.

Looking at the product yield of DEE decomposition, the main compounds are ethylene and ethanol, in a ratio not far from 1:1 within experimental error, in the low temperature range. This suggests that the reaction observed is:



A slight excess of ethanol with respect to ethylene could also be explained by the DEE hydrolysis reaction:



i.e. the inverse of reaction (2), taking into account that the DEE feed contains also small amounts of water. In fact alumina has been patented years ago as an excellent catalyst for reaction (4) [37].

At higher temperature, ethanol disappears among the products, showing that reaction (3) (DEE cracking) is followed by ethanol dehydration (reaction (1)). Thus DEE cracking can be performed producing ethylene with high yields. At high conversion, ethylene yields are lowered by the coproduction of ethane and higher hydrocarbons. As observed also upon ethanol dehydration, MFI zeolite can produce relevant amounts of aromatics thus significantly decreasing ethylene yields (from <90% to <75%). Also on alumina significant amounts of byproducts are obtained thus limiting ethylene yield to 91–93%. Higher yields to ethylene are obtained on USY (96.8% at 623 K) and on silica–alumina (>97% at >673 K).

It is actually relevant to remark that the temperature range for the start of DEE cracking reaction (3) corresponds to the temperature range where ethanol conversion to DEE (reaction (2)) is already significant and is starting to decline in favor of reaction (1) with ethanol feed. This suggests that reaction (3) can indeed be in sequence with reaction (2), producing reaction (1) with a consecutive path. However, it can be remarked that the feed was used in these DEE conversion experiments was definitely more concentrated than the feed was used in ethanol conversion experiments.

3.6. Conversion of ethanol + DEE + water mixtures

Another set of experiments was performed feeding a mixture of ethanol, DEE and water with a similar molar concentration used in ethanol conversion experiments (Fig. 8). Over alumina at 473 K, DEE and water are consumed and ethanol is formed, while ethylene is not observed. This indicates that, at this low temperature, alumina catalyses DEE hydrolysis (4), i.e. the inverse of reaction (2). As said, alumina has been patented years ago as an excellent catalyst for reaction (4) [38]. At 523 K, instead, the conversions of both DEE and water vanish and also ethanol conversion is near zero. This suggests that in these conditions reactions (2) and (4) very likely balance. At 573 K ethylene and water and some ethanol are produced, at the expense of DEE conversion. This shows that reactions (3) and (4) start. At 673 K, conversion of both DEE and

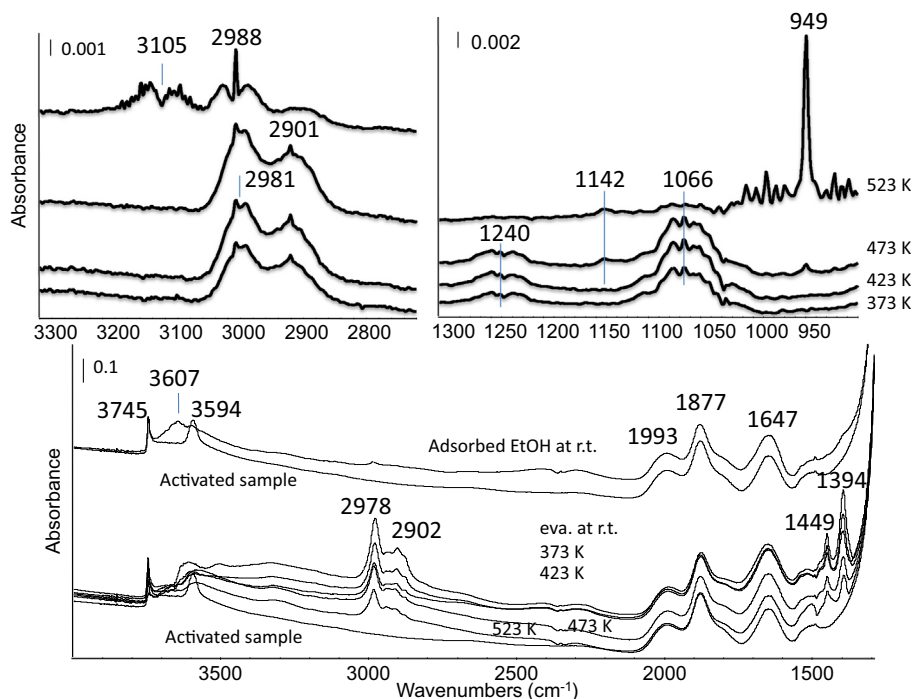


Fig. 6. FT-IR spectra of subtraction gas phase (top) and surface species (bottom) arising from ethanol adsorbed on FER.

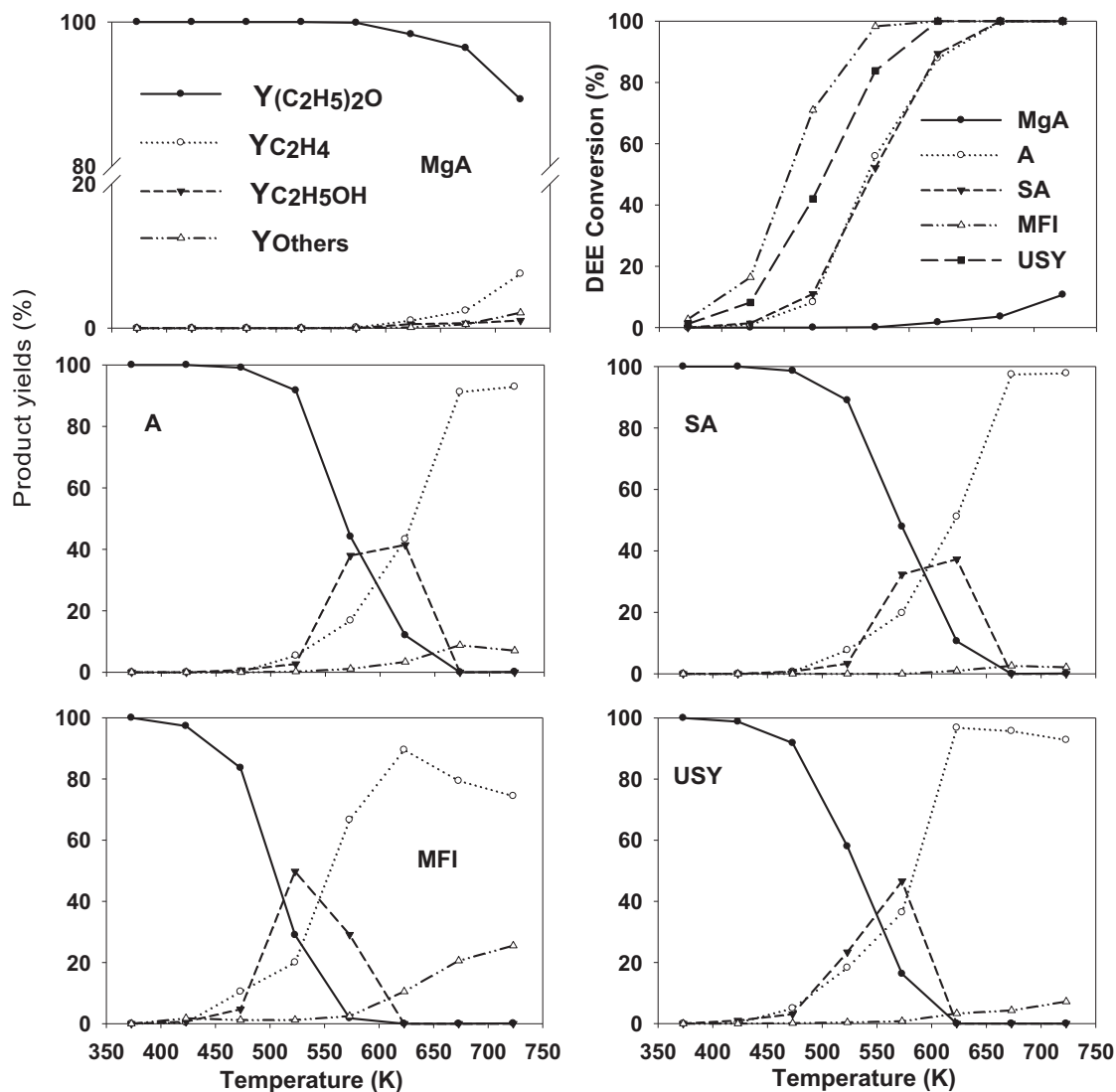


Fig. 7. Product yields over MgA, A, SA, MFI and USY catalysts at atmospheric pressure in a fixed-bed tubular quartz reactor using 0.5 g of catalysts with 10.42 h^{-1} WHSV in nitrogen, as a function of reaction temperature ($Y(C_2H_5)_2O = 100 - X(C_2H_5)_2O$). Right top: comparison of DEE conversions $X(C_2H_5)_2O$.

ethanol is complete with ethylene and water being the only products.

On MFI, ethanol conversion is higher than DEE conversion at 473 K, but is lower at 523 K, both producing ethylene only. Both are almost fully converted starting from 573 K. These data strongly support the idea that DEE can be intermediate in the ethanol dehydration reaction into ethylene: in fact, DEE cracks together with ethanol to form ethylene.

4. Discussion

The data described above show that over acidic catalysts both ethanol dehydration and DEE cracking are catalyzed efficiently. Zeolites are more active than alumina and silica-alumina in both cases, and silica-alumina is slightly less active than alumina in both reactions. Characterization data indicate, as it is well-known indeed, that zeolites are strong Brønsted acid catalysts while alumina mostly acts as a Lewis acid catalyst. The behavior of silica-alumina supports the idea that this material, although characterized also by Brønsted acidity sufficiently strong to protonate pyridine, likely works as a Lewis acid in these reactions. The

comparison of the behavior of silica-alumina with that of the MFI sample, characterized by a very low amount of Al and, thus, of Brønsted sites, further confirms that the active sites on silica-alumina are different from those of zeolites: in fact, the different behavior of MFI and silica-alumina cannot be attributed to the lower concentration of the same sites on SA.

The weakly acidic, or predominantly basic catalyst MgA (calcined hydrotalcite) is weakly active in both reactions, confirming the role of acid sites for them.

Upon ethanol conversion on all acidic catalysts, dehydration to DEE occurs with high selectivity at low temperature and conversion, while the reaction to ethylene occurs at higher temperature and conversion. Thermodynamic calculations indicate that, even in the conditions where DEE is formed with high selectivities, ethylene formation is thermodynamically far more favored. Thus thermodynamics is not controlling the product selectivity. In those conditions, the evaluation of the apparent activation energies in our experiments at low conversion ($>80 \text{ kJ/mol}$) reveals that the kinetic control is certainly chemical.

This result (DEE formation at low temperature and conversion; and ethylene formation at higher temperature and conversion) was observed also on less active and selective catalysts, but shifted to

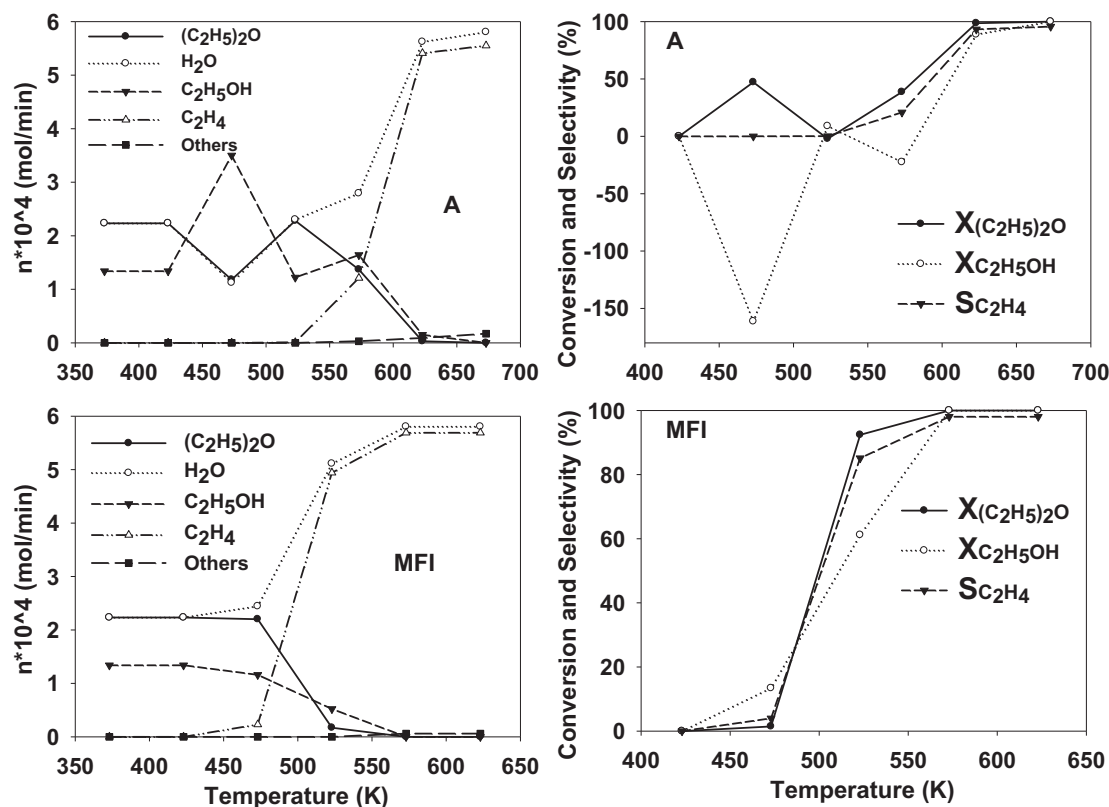


Fig. 8. Outlet molar flow of reactants and products (mol/min) (left); and conversions of diethyl ether and ethanol, and selectivity to ethylene (right) over 0.5 g of A and MFI catalysts at atmospheric pressure in a fixed-bed tubular quartz reactor feeding mixture of diethyl ether:water:ethanol = 5:5:3 (mol ratio in gas phase) in helium, as a function of reaction temperature.

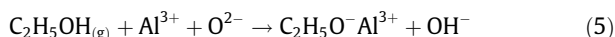
higher temperatures due to the lower activity of the catalysts. In these cases other compounds such as acetaldehyde, ethane and higher hydrocarbons may also be formed with significant selectivities [38].

Thus, the catalytic experiments show that at low temperature and conversion the reaction occurring in all cases with almost total selectivity is the dehydration of ethanol to DEE, reaction (2). This shows that, in these low temperature conditions, reaction (2) is much faster than reaction (1). At high temperature and conversion, instead, the reaction observed, also with high and almost total selectivity, is the dehydration of ethanol to ethylene (reaction (1)).

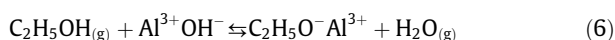
IR spectroscopy data confirm the central role of ethoxy groups both on alumina and on zeolites, both for producing DEE at low temperature and for producing ethylene at high temperature, in contrast with the conclusions of other authors [14,40].

As discussed previously [35], on alumina the formation of ethoxy groups occurs with two parallel mechanisms:

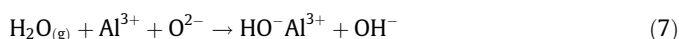
- (a) Dissociative adsorption on Lewis acid–base couples



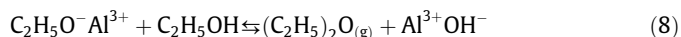
- (b) Exchange reaction on surface hydroxy groups



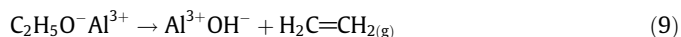
Both adsorption mechanisms undergo competition by the presence of water vapor. In fact, the latter is an invertible equilibrium being the acido-basic and volatility properties of water comparable to those of ethanol. The former competes with the dissociative adsorption of water on the same acid–base couples:



The inhibition of ethanol conversion by water does not need the existence of “dimeric” forms, supposed by other authors [14]. Ethoxy groups are stable on alumina below ca. 450–473 K. Starting from this temperature DEE is produced with very high selectivity and ethoxy groups are concomitantly consumed. This step is likely the following:

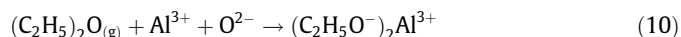


with only very small traces of ethylene coproduced. As confirmed by IR adsorption experiments, the adsorption of DEE is very weak and it easily desorbs. When ethanol gas is not more available and the temperature is higher, ethoxy groups decompose to gas phase ethylene and an hydroxyl group:



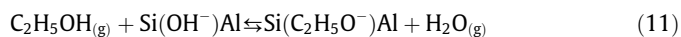
Ethylene adsorption is very weak on alumina, thus desorbing easily in the gas phase.

The DEE cracking reaction may likely occur either starting from the reverse of reaction (8) or by decomposition over acid–base couples:



followed by reaction (9) and by the reverse of reaction (5).

On zeolites the mechanisms are similar although the ethoxide group has a different geometry and is formed only by the exchange reaction:



Some authors suppose that reactions (1) and (2) are parallel [13,14,39] but few studies suggest that ethylene can be produced by a consecutive mechanism with reaction (2) followed by DEE cracking (reaction (3)) [12,17,18]. Our data suggest that indeed

both conclusions can be right. In fact, DEE cracking reaction (3) is fast over zeolite, alumina and silica–alumina catalysts in the conditions where DEE is formed by reaction (2). On the other hand, in agreement with several studies, the direct way to ethylene via an elimination reaction of ethoxide species is well substantiated to occur at high temperature over both alumina and zeolites by spectroscopic experiments, and by literature data.

We may mention that reaction (1) is expected to have a low reaction order with respect to ethanol (from one to negative depending on the ratio between adsorption strength and rate of surface reaction), while reaction (2) may have a larger reaction order (up to two) due to the need of two ethanol molecules to react together. This may justify an effect of ethanol concentration, and consequently of ethanol conversion, on the two reactions, favoring reaction (2) at high concentration and/or low conversion and reaction (1) at lower concentration and higher conversion. In agreement with this, Chiang and Bahn [39] found a positive reaction order of DEE formation with respect to ethanol on different zeolites, and a negative reaction order for ethylene formation from ethanol on H-MOR.

On the other hand, reaction (1), supposed to occur as a decomposition of ethoxy species, could have higher activation energy than reaction (2), thus also justifying it to become faster only at higher temperatures. In fact our data support the idea that dehydration to ethylene occurs with two paths, the direct one (through ethoxy groups cracking) and the consecutive one composed of formation of DEE by reaction of an ethoxy group with undissociated ethanol and successive cracking of DEE on the acid sites.

In summary, a likely reaction path on alumina catalyst is summarized in Scheme 1. Taking into account that water is a product of reactions 1 and 2 and that reaction temperature is low, it can be supposed that the surface is largely hydroxylated. Thus the active site can be supposed to be constituted by active hydroxyl groups as discussed previously [35]. Thus the first step, denoted as step A in Scheme 1, corresponds to reaction (6), producing ethoxy groups with desorption of water. Later, step B in the scheme gives rise to ethylene through a substantially irreversible elimination mechanism (reaction (9)). As a parallel way, in step C reaction with gas phase ethanol gives rise to diethyl ether in a reversible way. In fact, diethyl ether adsorption can crack with the sequence step –C, step –B producing ethylene too. As said, step B is faster than step C at high temperature and conversion while the reverse is true at

lower temperature and conversion. In the case of zeolite catalyst, the scheme of ethanol decomposition is essentially the same.

5. Conclusions

The main conclusions from these data are the following:

1. Acidic catalysts like zeolites, alumina and silica–alumina are active in the temperature range 453–573 K for both ethanol dehydration to DEE and to ethylene and for DEE cracking and hydrolysis, producing back ethanol and ethylene.
2. Protonic zeolites are more active than alumina which is slightly more active than silica–alumina for these reactions. It is confirmed that the active sites of silica–alumina are different and less active than those of protonic zeolites.
3. Ethanol dehydration to DEE occurs selectively at lower temperature with a “bimolecular” mechanism involving reaction of ethoxy groups with undissociated ethanol.
4. Ethanol dehydration to ethylene occurs selectively at relatively high temperature with an elimination mechanism via decomposition of ethoxy groups over these catalysts.
5. Ethanol dehydration to ethylene occurs also at lower temperature with a consecutive path via DEE formation and cracking.

Acknowledgements

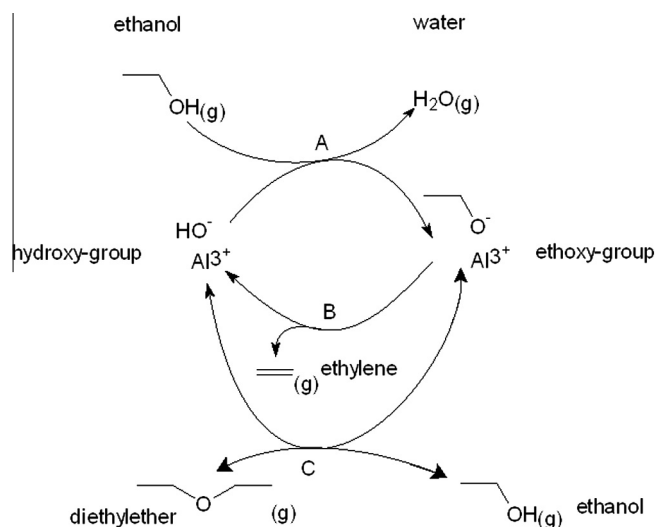
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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.cej.2015.03.008>.

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Scheme 1. Schematics of the catalytic cycles.

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